



Fourth Semester BE/B Tech Degree Examination: June/July 2026

Model Question Paper

Course: Discrete Mathematical Structures and Graph Theory (BCSEESC404A)

Maximum Marks: 100

Duration: 3 Hours

Instructions to the Students:

1. Answer any 5 full questions by choosing one full question from each module.
2. Mathematics Formula handbooks are allowed.

Q. No.	Question	Marks	RL	CO
Module 1				
1	a) Define the following with an example for each: i) Disjunction ii) Conjunction iii) Negation	6	L1	1
	b) Verify the compound proposition for tautology and contradiction $(p \vee q) \rightarrow r \leftrightarrow (\neg r \rightarrow \neg (p \vee q))$	7	L2	2
	c) Verify whether the following argument is valid. If a triangle has 2 equal sides, then it is isosceles. If the triangle is isosceles, then it has 2 equal angles. <u>A certain triangle ABC does not have 2 equal angles.</u> $\therefore$ the triangle ABC does not have 2 equal sides.	7	L2	2
OR				
2	a) State the converse, inverse and contrapositive of: i) If the triangle is not isosceles, then it is not equilateral ii) If the real number x^2 is greater than zero, then x is not equal to zero. iii) If a quadrilateral is a parallelogram, then its diagonals bisect each other.	6	L1	1
	b) Prove the logical equivalence $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$ using laws of logic.	7	L2	2
	c) Prove that, for all integers k and l , if k and l are both odd then $k + l$ is even and kl is odd.	7	L2	2
Module 2				
3	a) Make use of mathematical induction and prove that, for all positive integers $n \geq 1$, $1 + 2 + 3 + \dots + n = \frac{1}{2}n(n + 1)$	6	L2	2
	b) Determine the number of permutations of the letters of the word MASSASAUGA. In how many of these, all four 'A's are together? How many of them begin with S?	7	L3	2
	c) A committee of 12 is to be selected from 10 men and 10 women. Determine in how many ways can the selection be carried out if (a) There are no restrictions (b) There must be six men and six women (c) There must be an even number of women	7	L3	2
OR				



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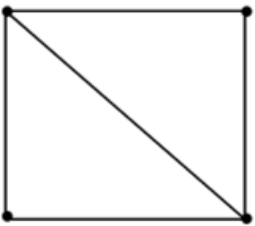
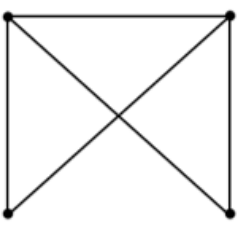
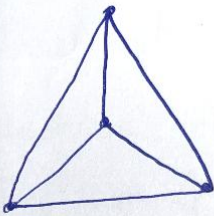
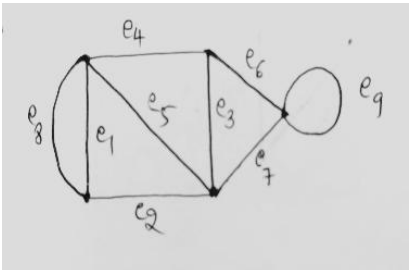
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Q. No.		Question	Marks	RL	CO									
4	a)	Make use of mathematical induction and prove that, for every positive integer n , 5 divides $n^5 - n$	6	L2	2									
	b)	Determine the coefficient of (i) xyz^2 in the expansion of $(2x - y - z)^4$ (ii) $x^{11}y^4$ in the expansion of $(2x^3 - 3xy^2 + z^2)^6$ (iii) x^{12} in the expansion of $x^3(1 - 2x)^{10}$	7	L3	2									
	c)	Determine in how many ways can 10 identical pencils be distributed among 5 children in the following cases: i) There are no restrictions. ii) Each child gets atleast one pencil. iii) The youngest child gets atleast two pencils.	7	L3	2									
Module 3														
5	a)	Determine in how many ways can one arrange the letters in the word CORRESPONDENTS so that (i) There is no pair of consecutive identical letters (ii) There are exactly two pairs of consecutive identical letters (iii) There are atleast three pairs of consecutive identical letters	6	L3	3									
	b)	Determine the rook polynomial for the board shown below <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td></td><td>5</td></tr><tr><td>6</td><td>7</td><td>8</td></tr></table>	1	2	3	4		5	6	7	8	7	L3	3
	1	2	3											
4		5												
6	7	8												
c)	The number of virus affected files in a system is 1000(to start with) and this increases 250% every two hours. Use a recurrence relation to determine the number of virus affected files in the system after one day.	7	L3	5										
OR														
6	a)	Solve the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 2$, given that $a_1 = 5, a_2 = 3$	6	L3	3									
	b)	Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 one teacher per each class. T_1 and T_2 do not wish to become the class teacher for C_1 or C_2 , T_3 and T_4 for C_4 or C_5 and, T_5 for C_3 or C_4 or C_5 . Determine in how many ways can the teachers be assigned the work. (Without displacing any teacher)	7	L3	3									
	c)	A bank pays a certain percent of annual interest on deposits compounding the interest once in three months. If a deposit doubles in 6 years and 6 months, determine what is the annual percent of interest paid by the bank.	7	L3	5									
Module 4														
7	a)	Define i) Regular graph ii) Trivial graph iii) Null graph with an example for each.	6	L1	1									
	b)	Analyse the difference between Complete graph and Complete Bipartite graph.	7	L4	1									
	c)	Construct two Euler graphs and an Unicursal graph by specifying the walk.	7	L3	4									



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OR				
8	a) Define i) Isomorphism ii) Euler graph iii) Hamilton path with an example for each.	6	L1	1
	b) Analyse whether the following graphs are isomorphic or not.   G_1 G_2	7	L4	1
	c) Explain Hamilton cycle with an example. Determine the number of edge-disjoint Hamilton cycles exist in a complete graph with 5 vertices. Hence construct the graph to show and specify the cycle.	7	L3	4
Module 5				
9	a) Show that a connected graph G with n vertices and e edges have $e - n + 2$ regions	6	L2	4
	b) Construct a graph for both planar graph and non-planar graph and give two conditions for testing planarity of any graph.	7	L3	4
	c) Construct the geometric dual of the following graph. 	7	L3	4
OR				
10	a) Explain the types of digraphs with an example for each.	6	L2	4
	b) Investigate for planarity of the following graph by the method of elementary reduction 	7	L3	4



Model Question Paper

Q. No.		Question	Marks	RL	CO
	c)	Determine the adjacency matrix and incidence matrix for the given graph.	7	L3	4

Paper Setter

DAAC

Chairman